Forest Fire Early Warning Sensor Network: A Discrete-Event Simulation Study Using OMNET++, INET, and FLoRa

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GitHub:https://github.com/alpkaanozgul/Forest-Fire-Early-Warning-Sensor-Network

***Abstract*—This project presents the design, modeling, and evaluation of a Forest Fire Early Warning Sensor Network simulation. The system deploys smoke, temperature, humidity, and wind-speed sensors across a forested region, communicating through LoRa/LoRaWAN gateways to a central monitoring station. The project developed in OMNET++(OMNET++ 6.0.3) application with using INET(INET 4.4) for backend network and Flora(Flora 1.1.0) for lora communication modelling. This project mainly demonstrates forest fire warning sensor network, under different traffic conditions, varying sensor densities and environmental factors with using consecutive Monte Carlo simulations and statistical input and output analysis. In the implementation part our team were strictly followed to our interim report and CNG 476 course materials. All the related documentation, source codes and analysis scripts are shared via GitHub.**

***Index Terms*—System simulation, OMNET++, INET, FLoRa, IoT, discrete-event simulation, forest fire detection.**

I. Introduction

Forest fires are severe threat to ecosystems, human beings, and economic assets. Therefore, early detection of fire events is critical to minimising damage, yet remote forested areas are often inaccessible for continuous human activities. Internet of Things (IoT) sensor networks offer a good alternative: autonomous, low-power devices can continuously monitor environmental parameters and transmit warnings over long range, low-power wireless links.

This project proposes in which smoke detectors, temperature sensors, humidity sensors, and wind-speed sensors are distributed across a forest area. These sensor nodes communicate via LoRa/LoRaWAN technology through one or more gateways to a central fire monitoring server. Under normal conditions, nodes transmit periodic telemetry reports at low duty cycles. When sensor readings exceed predefined fire risk thresholds, alarm messages are generated and send to the monitoring system.

The simulation developed in the OMNET++ as an discrete event simulator.INET for IP-based backend communication between gateways and the monitoring server, and FLoRa for modelling LoRa physical-layer and LoRaWAN MAC-layer behaviour. For the OMNET++ both WSL and virual machine systems were used in implementation as a workspace. In this report, previous expectations, simulation results, implementation phase, course materials and the requirements from the interim reports will be discussed in detail.

II. Selected Project Description

Forest Fire Early Warning Sensor Network system mainly distributes smoke detectors, temperature sensors, humidity sensors, and wind sensors across a forest. For simulating real life conditions in the OMNET++ system. Under normal conditions, nodes sends periodic signals. When sensor readings exceed predefined fire-risk thresholds, high-priority alarm messages will be send. The detailed information about Forest Fire Early Warning Sensor Network simulation project is given in Table 1.

TABLE I
Selected Project Summary

Item	Description
Selected Project Title	Forest Fire Early Warning Sensor Network
Application Domain	Internet of Things (IoT) / Environmental Monitoring
Main Problem	Gateway congestion and wireless collisions during simultaneous multi-sensor fire alarm under heavy stress
Simulation Objective	Evaluate end-to-end delay and packet delivery ratio under varying network stress and reporting policies
Main Users/Entities	Sensor Nodes, LoRa Gateways, Fire Monitoring Server, Fire Department
Main Performance Metrics	End-to-End Alarm Delay, Packet Delivery Ratio (PDR), Gateway Queue Waiting Time

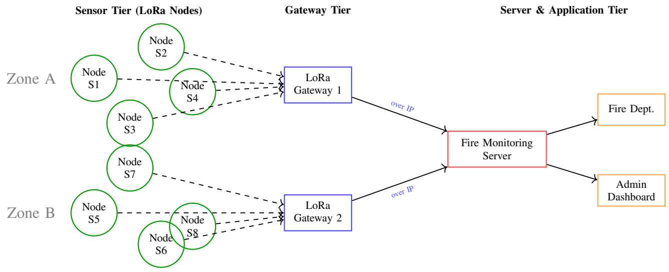

Fig. 1. System architecture of the Forest Fire Early Warning Sensor Network showing the three-tier IoT topology: 8 different nodes each four of them grouped as an one zone and linked to one gateway. Then, each gateway for specified zone is connected to main Fire Monitoring Server, which is the computational force of the system.

III. System Architecture

The system follows a three-tier IoT topology consisting of a Sensor Tier (LoRa Nodes) which includes 8 nodes at total grouped four by four for each zone, a Gateway Tier also for each zone one gateway tier is used, and a Server & Application Tier.

The physical deployment and network mechanics operate through the following structured layers:

- **Sensor Tier (LoRa Nodes):** Eight specialized sensor nodes are organized into two zones, named as Zone A (containing sensors S1 through S4) and Zone B (containing sensors S5 through S8). These nodes are responsible for collecting data from the field, including smoke densities, temperature shifts, regional humidity percentages, and wind-speed data. Then sends them to their own zones gateway tier.
- **Gateway Tier:** For each zone one gateway tier is deployed for collecting telemetry data from the linked nodes in their own zones and pass it to the central Fire Monitoring Server.
- **Server & Application Tier:** Serving as the final data processing destination, the central Fire Monitoring Server is linked directly to the LoRa gateways through standard IP networking provided via INET simulation layers. Then sends final package to admin side and Fire Department.

In summary, sensor tier nodes collect data and transmits them via LoRa to gateways, which then forward data over IP using INET to the central fire monitoring server. The server sends signals to the fire department and admin panel.

IV. System Model and Assumptions

Our simulation model is built over four main entities. The first one is sensor node, this entity is responsible for collecting and transmitting telemetry data from the assigned zone and detecting alarm conditions. The second entity is LoRa channel which is mainly responsible for transmitting data to its assigned LoRa gateway successfully. The third entity, LoRa gateway is responsible for receiving telemetry data from the nodes via LoRa channel and queue handling process in the high densities or concurrent time alarms therefore it plays very critical role in our simulation model. The last main entity is the monitoring server, it represents the main computational power of our model and the last step before the fire department. All the related information about system model is also provided in Table 2.

For this system, in the design phase we made some assumptions for facilitate the simulation process. Our first assumption is that our sensors will be distributed in the flat area and their positions stay same during the simulation to neglect topological errors. The second assumption is log distance path loss, because we know that in real world as the interval between transmitter and receiver increases, signal power decreases logarithmically so we have an assumption that FloRa is realistic and reliable model for simulating this issue. Our third assumption is Single-Channel Gateway Reception per SF, this restriction is very important because without assuming this restriction every alarms can works perfectly under all conditions which is not the point of simulations. Our last assumption is delay caused by network communication is fixed for all packages between LoRa gateways and main monitoring server. All this assumptions are also concluded in Table 3.

TABLE II
Simulation Entities, Events, and State Variables

Entity	Events	State Variables
Sensor Node	Telemetry generation, Alarm generation, Timeout	Operational mode (idle, sensing, transmitting, alarm)
LoRa Channel	Collision event	Channel status (idle, busy, collision)
LoRa Gateway	Packet arrival, Service start, Service completion	Queue status, buffer occupancy
Monitoring Server	Packet delivery to server	Processed alarm count, system throughput

TABLE III
Modelling Assumptions and Justification

Assumption	Justification
Stationary sensors in a 2D flat area	Simplifies the node deployment and spatial coordinates for radio propagation mapping.
Log-distance path loss model	Standard and reliable analytical model for urban/forest wireless propagation in FLoRa.
Single-channel gateway reception per SF	Reflects hardware limits where concurrent packets on the same SF cause collisions.
Fixed delay for backend IP links	Simplifies backend network behavior to focus bottleneck analysis on the wireless LoRa tier.

V. Simulation Methodology

The project employs a Discrete-Event Simulation (DES) methodology. The state of the system changes only after particular events, these events changes the state of components like nodes or queue. To capture the stochastic nature of fire ignitions and channel contentions this simulation executed as a Monte Carlo experiment and it is implemented within the component-based architecture of OMNeT++. Also, different RNG streams are used for different nodes and chanels to prevent a correlation between different nodes. All stochastic processes and environmental variables are modelled with using CNG476 course materials. In the Table 4, each process and their distribution model choice and the reason behind them is explained in detail. Also in Table 5, trigger conditions for each specified events and the their possible results are given in order.

TABLE IV
RANDOM PROCESSES AND PROBABILITY MODELS

Process	**Distribution**	**Purpose in Simulation**
Packet generation	Exponential (λ)	Inter-transmission times of periodic telemetry
Sensing event	Normal (μ, σ^2)	Ambient temperature and humidity readings
Failure/alarm event	Bernoulli (p) / Poisson	False alarm probability; Number of fire events
Service time	Exponential (μ)	Processing delay at the gateway queue
Back-off delay	Uniform / Geometric	Collision back-off slots and transmission attempts

TABLE V
SCHEDULED EVENTS IN THE DISCRETE-EVENT SIMULATION

Event	**Trigger Condition**	**State Change / Action**
TelemetryGen.	Periodic timer expiry	Node creates telemetry packet and sends them to gateway.
FireIgnition	Global Poisson trigger	Generates fire point, changes environment variables.
AlarmGen.	Reading exceeds limit values	Node switches to alarm mode, sends priority packet.
ServiceStart	Gateway queue item ready	Gateway server becomes busy, processes packet.
CollisionBackoff	Packet collision occurs	Node backs off for a random interval before another try.
TimeoutEvent	Nothing happens for long time	Triggers packet retransmission attempt.

VI. OMNET++, INET, AND FLORA IMPLEMENTATION

This part underlines the details of the implementation architecture of the project within the OMNET++ DES environment as the simulation engine, using the INET frameworks which provides the basic network infrastructure and FLoRa framework libraries and extensions, which are responsible for LoRaWAN MAC and physical layer modelling. The project simulates a network of LoRa-equipped sensor nodes distributed throughout a forest area, designed to measure environmental parameters and transmit alerts to a remote monitoring station via a LoRaWAN Gateway and Network Server.

In this project we implemented 4 different ned files for each of the components consecutively; Fire, Gateway, Sensor and Server modules. All these files are creates simulations component-oriented structure, each one of them implements their modules external parameter interface and places in the main simulation.

In addition, for each of these four modules we implemented source codes in c++ environment. For each component these part consist of 2 files. The first one is header file for each of the component, these files are responsible for providing required libraries and informations for their own modules source codes for example relation between fireGen.h and fireGen.cc. Also for combining each components in the main simulation and run them properly we have an ”utils” file which consist of RngUtils.h and RngUtils.cc. For the detailed information for each of these files are explained in Table 6. Also in Table 7, OMNET++, INET and FloRa tools and frameworks are explained consecutively about how they are used in the project.

In the project, under the msg folder we have two distinc message files which are representing the messages send by individual nodes. The firest message file is ”AlarmMsg.msg” file this contains the msg data sent by nodes to the gateways under the real fire events. The second msg file is ”TelemetryMsg.msg”, this file contains regular telemetry data like humidity or temperature in the specifed node in the zone, as explained in detail in the other sections these telemetry messages sends regularly from the nodes not only in the real fire conditions.

In our project, omnetpp.ini files mainly operates simulation parameters and different test scenerios in our architecture consisting of Sensor Nodes, $M/M/1/K$ Gateway Queues, and a main Server. In this configuration each scenario runs 30 independent times, at max 3600 seconds. Also, for each of the scenarios simulation first tries to reach steady state with 300 second warm-up period. The file defines five distinct modes to evaluate system: Baseline, DenseNodes (node density increased), MultiGateway (gateway sweep for redundancy), Under high fire stress and Fixed/Adaptive Interval (comparison of telemetry reporting frequencies).

After the project execution, the log datas taken from OMNET++ DES tool are provided in Figure 2 (These lines are selected randomly among the simulation output logs in OMNET++)

VII. MAPPING OF COURSE TOPICS TO THE PROJECT

In this section how the each of the course materials from CNG476, will be shown and explained in detail. While designing and implementing processes we tried to stick in to the related materials.Lecture slides and notes were actively used during development. For each of the course topic and how

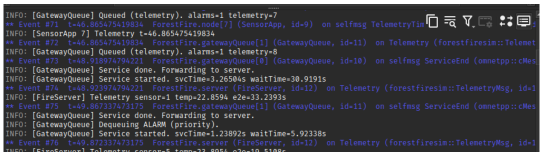

Fig. 2. Log results from OMNET++

TABLE VI
IMPLEMENTED OR MODIFIED PROJECT FILES

File Name	File Type	Purpose
Fire	NED	Structure & area config
Fire	Header	Dynamic simulation declarations
Fire	C++	Fire spreading execution
Gateway	NED	Module layout & gates
Gateway	Header	Data buffer & routing table
Gateway	C++	Data aggregation & forwarding
Sensor	NED	Node parameters & gates
Sensor	Header	Mechanisms & thresholds
Sensor	C++	Sampling & alert packaging
Server	NED	Server config & ports
Server	Header	Loggers & alert queues
Server	C++	Processing & alerts evaluation
Utils	NED	Global configurations
Utils	Header	Common utility functions
Utils	C++	Calculations & formatting

they used in simulation and development process is shown in Table 8.

VIII. EXPERIMENTAL DESIGN

In our simulation, we designed the system in a way that we will be able to test M/M/1 queue structure in 3 layered IoT module and we made all the configurations in the omnetpp.ini file. For the statistical requirements, each of the simulation scenarios were repeated 30 times with random seeds and with having 300 seconds of warm-up time system simulated in 3600 seconds. During the simulation we simulated 5 different scenario for simulating different real world scenarios. In our baseline scenario we have 8 nodes, 2 gateways and 1 central server and regular conditions but in the other modes for simulating different scenarios we implemented a DenseNodes,MultiGateway, HighFireRate and Fixed/Adaptive interval mode. Table 10 shows the definition of each metrics with their units.

TABLE VII
OMNET++, INET, AND FLORA USAGE

Tool/Framework	How It Is Used in the Project
OMNET++	Core discrete-event engine and module architecture setup via NED.
INET	Models IP backend links, routing, and propagation delay.
FLoRa	Simulates LoRa physical/MAC behavior, SFs, and collisions.

TABLE VIII
MAPPING OF CNG 476 COURSE TOPICS TO THE FINAL PROJECT

Course Topic	Project Implementation
Sim. methodology	Conceptual design to M/M/1 validation.
Random numbers	Uniform(0,1) isolation per module.
Monte Carlo	30 independent replications per profile.
Probability models	Exponential and Poisson arrival maps.
Inverse transform	CDF-based generation in C++ code.
Stochastic processes	Poisson and exponential capacities.
Queueing analysis	Gateway buffer as $M/M/1/K$ system.
Discrete-event sim	Event-triggered state transitions.
Event scheduling	Pre-scheduled sensor telemetry timers.
Time-stepped views	Environmental updates & weather feeds.
Wireless network	LoRa capture effect and collisions.
OMNET++ platform	Custom simple modules and network files.
INET integration	Gateway-to-server IP routing tier.
FLoRa integration	LoRaWAN MAC prioritization stress.
Input analysis	Histograms and QQ-plot dataset checks.
Output analysis	Welch method and confidence intervals.
IoT relevance	LPWAN forest fire safety network issues.

TABLE IX
EXPERIMENTAL DESIGN SUMMARY

Item	Value / Description
Simulation Duration	3600 seconds (1 Hour) per run
Warm-up Period	300 s (discarded for steady-state)
Number of Runs	30 runs (`repeat = 30`)
Random Seeds	`seed-set = $repetition`
Baseline Setup	$N = 8$, $G = 2$, $\lambda_{fire} = 0.005$, $T_{mean} = 30\text{s}$
Alternative Configs	DenseNodes, MultiGateway, HighFireRate, Interval sweeps
Varied Parameters	$N \in \{4..20\}$, $G \in \{1..3\}$, $\lambda \in \{0.001..0.02\}$, $T \in \{15\text{s}, 30\text{s}\}$
Collected Metrics	Delay, loss, throughput, drop statistics

IX. PERFORMANCE METRICS

In our simulation project to be able to evaluate the simulations consistency, performance and reliability.For these performance evaluation we kept record of End to end delay, delivery ratio for each package, packet drop rate, waiting time in the queue and the throughput of the system. These metrics are shown in detail at the Table 10 with their units and how and what they used for.

X. INPUT ANALYSIS

In our simulation to be able to guarantee our random input's validity we performed histograms and Q&Q plots with using emprical input values.For the sensor temperature values

TABLE X
PERFORMANCE METRICS

Metric Name	Unit	Definition / Purpose
End-to-End Delay	s	Total time from packet generation by a sensor until server reception. Measures alert latency.
Packet Delivery Ratio (PDR)	%	Ratio of packets received at the server to total transmitted packets. Measures network reliability.
Packet Drop Rate	%	Discarded packets at Gateway due to $M/M/1/K$ buffer overflow. Tracks congestion.
Queue Waiting Time	s	Average buffer wait time before service. Detects bottlenecks.
Throughput	pkts/s	Rate of successful packet delivery per unit of time. Tracks network capacity.

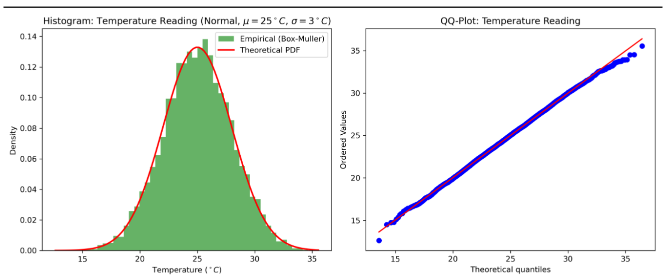

Fig. 3. Input Analysis: Baseline Temperature Distribution and QQ-Plot Fit

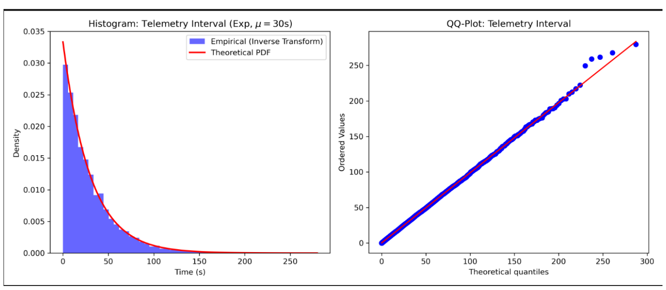

Fig. 4. Input Analysis: Telemetry Interval Distribution and QQ-Plot Fit

we have a Q&Q plot which match its reference points and validate our temperature data follow the normal distribution. In addition, our Q&Q plot for interval of sensor telemetry data validates its own standards by referencing exponential distribution standards. Table 11 shows the methods utilized to validate the parameters and their conclusions in detail.

XI. RESULTS AND OUTPUT ANALYSIS

To be able to evaluate and understand the result of our simulation, we implemented a five scenarios in omnet++ and simulated for 3600 seconds, includes 30 Monte Carlo experiment. In Table 12, for each of the scenarios metrics, avarage values, and standard deviation or confidence interval are given and explained in detail. All the given data in the table was derived from the simulation results and their graphs. These graphs can be seen in Figure 5, Figure 6 and Figure 7 consecutively.

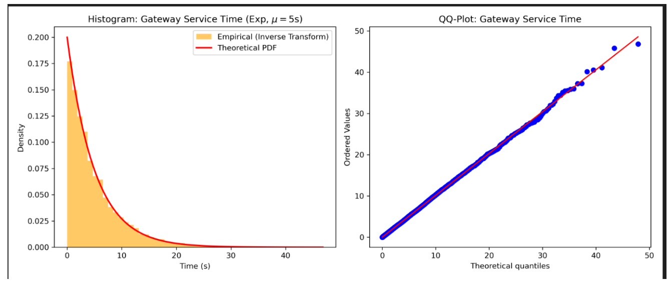

Fig. 5. Input Analysis: Gateway Service Time Distribution and QQ-Plot Fit

TABLE XI
INPUT ANALYSIS SUMMARY

Input Variable	Method	Conclusion
Baseline Temp.	Hist. & QQ	Fits Normal distribution ($\mu = 25°C, \sigma = 3°C$) via Box-Muller tracking.
Telemetry Interval	Hist. & QQ	Fits Exponential distribution ($\mu = 30s$) via Inverse Transform.
Gateway Service	Hist. & QQ	Fits Exponential distribution ($\mu = 5s$) verifying $M/M/1/K$ rules.

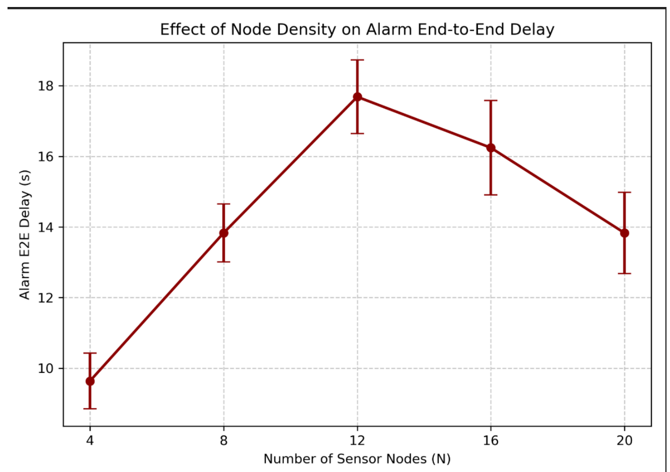

Fig. 6. Effect of Node Density on Alarm End-to-End Delay

XII. DISCUSSION

The result of our simulation shows that as long as the we increased number of nodes in the area we will able to cover greater zones but at the same time it causes to increase stress over the gateways and main server or collisions. In the HighFireRate scenario, system strugles to handle queue overflows and some packages might drop during the execution but when we implemented MultiGateway scenerio our simulation more easily handles this system bottlenecks. In general we were able to see that our simulation model's emprical result in the

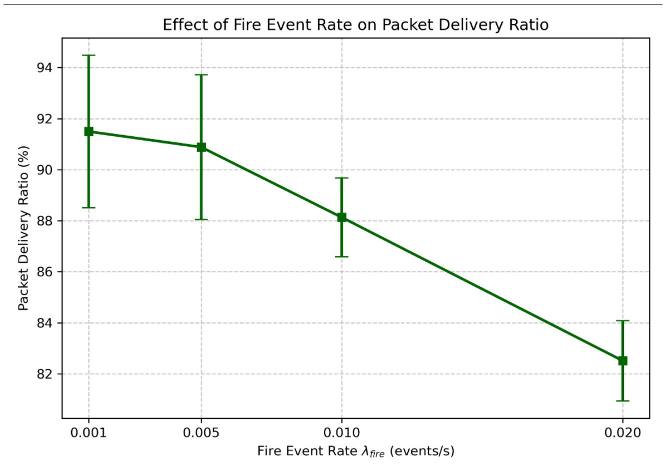

Fig. 7. Effect of Fire Event Rate on Packet Delivery Ratio

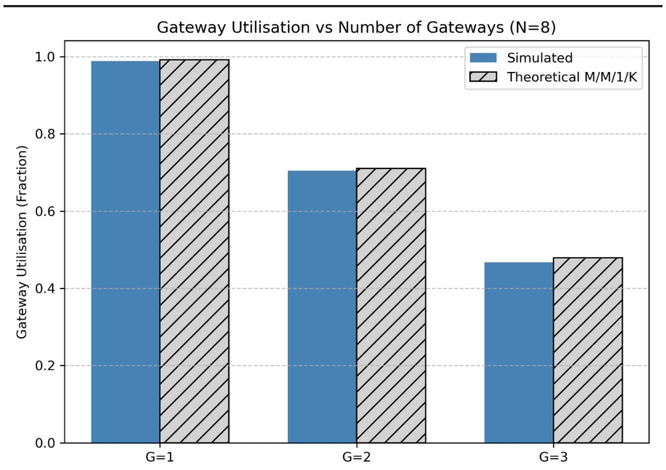

Fig. 8. Gateway Utilisation vs Number of Gateways ($N = 8$)

simulations steady state was able to validate structural and queue perspective of theoretical M/M/1/K model. The detailed findings for each of the scenario and their explanation is given in Table 13.

XIII. Conclusion

In the Forest Fire Warning Sensor simulating project, we implemented a discrete time event simulation model in OMNET++ and actively used INET(4.4) and FloRa (1.1.0)

TABLE XII
Simulation Results Summary

Scenario / Mode	Metric	Avg	Std. Dev.	CI (90%)
Baseline	Gateway Utilisation (ρ) Packet Delivery Ratio (PDR)	0.704 90.90%	0.015 1.35%	$\pm$ 0.005 $\pm$ 0.41%
DenseNodes	Alarm E2E Delay (D_{alarm})	16.25 s	4.35 s	$\pm$ 1.32 s
MultiGateway	Gateway Utilisation (ρ)	0.468	0.021	$\pm$ 0.006
HighFireRate	Packet Delivery Ratio (PDR)	82.52%	1.55%	$\pm$ 0.47%
Fixed/Adaptive	Alarm E2E Delay (D_{alarm})	13.84 s	3.79 s	$\pm$ 1.15 s

TABLE XIII
Discussion of Main Findings

Scenario / Mode	Main Finding & Technical Explanation
Baseline	It represents normal conditions ($N = 8$, $G = 2$). With high delivery ratio (90.9%) and steady gateway load (0.70), it satisfied our expectations from therotical lectures.
DenseNodes	Non-monotonic delay trend. More nodes mean more packet collisions at first, but very high density creates shorter routing paths that lower the delay.
HighFireRate	When we increased number of real alarms to the gateways, strugle to queue and causes some package drops. These reasons caused to delivery ratio to down.
MultiGateway	Lower gateway load (0.468). Adding more gateways ($G = 3$) shares the traffic load and removes network bottlenecks. Therefore it helped the system in positive manner.
Fixed/Adaptive	Balanced traffic. Mixing regular updates with alarm triggers creates steady traffic that follows standard queue models.

frameworks, while advancing simulation methodologies also review the CNG 476 course materials. The performance metrics showed us multi-gateway approaches offers a better and safer solutions for IoT applications. The most important limitation in the design process was deciding about which simulation scenarios we should implement and analyze for having valid outcome. As an future work dynamic wind speed integration will be considered because it can helpful for advancing simulation.

Code Availability

All of the related materials about simulation(source codes, simulation, message and message files) and reports are listed in our GitHub repository: https://github.com/alpkaanozgul/Forest-Fire-Early-Warning-Sensor-Network

Individual Contributions

As we mentioned in the our interim report we mainly worked together for coding, designing and simulation result analysis part. Also, detailed role distrubition given in Table 14.

TABLE XIV
Individual Contributions

Name	ID	Contribution
Alp Kaan Özgül	2638096	Environment setup, FLoRa integration, random variate generators, input analysis, code implementation, and final report.
Yasin Bora Buğdacı	2584753	Application logic, gateway queue module, output processing, code implementation, and final report.

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